PROJECT REPORT

TECHNOLOGY INNOVATIONS AND COMPETITIVE ADVANTAGE: A STUDY OF TESLA

A secondary-data comparative case study Tesla financial analysis 2019-2025 Comparative context: BYD and Toyota

Submitted By:

Lakshya Sharma 23111018

Samar Bansal 23111004

Anirudh Sharma 23111016

Suryadipta Ghosh 23111002

Jashan Bansal 23104021

Akshat Goyal 23111014

Under the guidance of:

Dr. Nidhi Tanwar

Assistant Professor

Department of Centre for Management and Humanities

Punjab Engineering College (Deemed to be University), Chandigarh

CONTENTS

1. Abstract 3

2. Introduction 4

3. Review of Literature 6

4. Need of the Study and Research Gap 10

5. Research Methodology 11

6. Data Analysis and Interpretation 14

7. Recommendations and Implications 21

8. Conclusion 22

9. Bibliography 23

10. Appendix 24

Figures and tables

Three line graphs, two bar graphs, and two pie charts show the financial trends and the 2025 business mix. Tables summarise the literature, reported data, calculations, and strategic assessment.

1. ABSTRACT

This study examines how Tesla's technological innovations contribute to competitive advantage and whether that advantage is visible in financial performance. It uses a descriptive and analytical case-study design based on seven annual observations from 2019 to 2025, company disclosures, and ten recent research studies. BYD and Toyota provide contextual comparisons, with differences in accounting definitions and reporting periods retained. The analysis combines financial ratios, growth rates, exploratory correlations, regulatory-credit sensitivity, and strategic frameworks. Tesla's revenue increased from $24.58 billion to $94.83 billion, while research and development expenditure rose from $1.34 billion to $6.41 billion. However, gross margin fell from 25.60% in 2022 to 18.03% in 2025. The contemporaneous association between R&D intensity and gross margin was negative, but the small sample and reversal in the lagged association prevent a causal interpretation. Software integration, charging infrastructure, and manufacturing knowledge offer useful advantages, although their durability depends on imitation, customer experience, and execution. Energy storage adds a further source of value. The study concludes that Tesla's advantage rests on coordinating complementary capabilities rather than on innovation expenditure alone. It recommends clearer project milestones, disciplined manufacturing investment, reliable charging services, and evaluation of commercial outcomes alongside technical progress.

Keywords: Tesla, technological innovation, competitive advantage, electric vehicles, R&D, charging infrastructure, energy storage

2. INTRODUCTION

2.1 Background of the topic

Electric vehicles have changed the basis of competition in the automobile industry. Battery performance, power electronics, software, and charging access now influence the ownership experience alongside price, reliability, and design. The International Energy Agency (IEA, 2025) reports that global electric car sales exceeded 17 million in 2024, accounting for more than one-fifth of car sales. Its definition includes battery electric and plug-in hybrid cars; it should not be confused with Tesla's battery-only vehicle business.

A growing industry does not guarantee stronger profitability for every producer. When competitors improve their products and production capacity expands, buyers gain alternatives. Some of the value created by innovation may pass to customers through lower prices. A useful study must therefore consider both technological capability and the firm's ability to retain an economic return.

2.2 Industry and company profile

Tesla was incorporated in 2003 and operates across electric vehicles, related services, and energy generation and storage. Its products combine physical engineering with software, while its charging network supports vehicle use beyond the initial sale. Powerwall and Megapack extend battery capabilities into stationary storage (Tesla, 2021, 2026). This combination makes Tesla a suitable case for studying innovation across connected activities.

BYD and Toyota represent different competitive paths. BYD provides a comparison with another large electrified-vehicle and battery business. Toyota brings the perspective of an established manufacturer with a broad powertrain portfolio. They are contextual comparators, not a statistically matched control group. Differences in business mix, accounting, and fiscal calendars limit direct comparisons.

2.3 Problem statement and scope

The central problem is whether Tesla's technological innovations create a defensible advantage that remains visible as the market matures. Revenue expansion may reflect scale, pricing, or market growth rather than a durable technological lead. Equally, rising R&D spending may support future projects whose returns are not yet observable. This report asks how the mechanisms connecting innovation to performance operate, where the evidence supports them, and where the claims remain uncertain.

The quantitative study covers Tesla's calendar years 2019-2025. Sources available by 14 October 2026 are used, including Toyota's year ended March 2026 for context. Tesla's 2026 interim results are excluded to preserve a consistent annual series.

2.4 Approach to competitive advantage

The study connects technological capabilities with customer value and financial outcomes. The analysis follows three questions: what the innovation changes, how that change could benefit customers or production, and whether Tesla can retain an advantage as competitors respond. The financial ratios are interpreted alongside those mechanisms.

Revenue growth is considered together with gross margin, R&D intensity, the contribution of regulatory credits, and the role of energy storage. This approach makes room for mixed results: a firm can expand its activities and improve its products while facing pressure on the financial returns it retains.

3. REVIEW OF LITERATURE

Author and year	Context and method	Main finding
Ziegler and Trancik (2021)	Battery price histories	Measure cost alongside performance.
Mauler et al. (2021)	53 forecasting studies	Forecasts depend on assumptions.
Yao et al. (2025)	6,000-plus scenarios	Chemistry gains depend on execution.
Stockkamp et al. (2021)	49 adoption studies	Adoption depends on several factors.
Hardman and Tal (2021)	California owner surveys	Charging supports continued ownership.
Springel (2021)	Norway market model	Charging and vehicle demand interact.
Powell et al. (2022)	Grid-dispatch modelling	Charging timing affects grid needs.
Alochet et al. (2023)	19-automaker comparison	Production remains partly integrated.
Wang et al. (2023)	China material scenarios	Material costs can slow adoption.
Hanig et al. (2025)	US coverage analysis	Counts alone miss charging gaps.

Table 1: Summary of recent studies reviewed

The review covers ten studies published during 2021-2025. Foundational strategy works are used separately and do not count toward this total. The following discussion retains each study's method, relevance to Tesla, and limitations.

3.1 Battery improvement and the economics of innovation

Study 1 - Ziegler and Trancik (2021). The authors reconcile historical price and performance series to examine lithium-ion battery improvement. Their analysis shows why cost comparisons should account for changes in the service a battery provides, including energy density. For Tesla, this supports evaluating usable range, weight, and cost together. The evidence is industry-wide, however, and cannot establish that Tesla owns the gains or earns a premium from them. Historical learning also does not guarantee the same future rate of improvement.

Study 2 - Mauler et al. (2021). This review examines 53 battery-cost forecasting studies and distinguishes technological learning, literature projections, expert judgments, and bottom-up models. Forecasts vary with their assumptions and treatment of technology. The study is useful because it discourages treating a single forecast as a business outcome. Tesla's battery investments should be judged against measured production costs and yields. The review estimates possible cost trajectories; it does not verify Tesla's cell-level economics or the commercial success of a particular manufacturing process.

Study 3 - Yao et al. (2025). Using component-level learning models and more than 6,000 scenarios, the authors evaluate sodium-ion batteries against lithium-ion alternatives. They find that near-term cost superiority is difficult and depends on energy density, engineering progress, and mineral prices. The implication is that a new chemistry is an option to evaluate, not an automatic source of advantage. For Tesla, flexibility across battery applications may matter more than committing every product to

one technology. These are conditional model results, not observations of mass-market sodium-ion profitability.

Interpretation

Together, these studies suggest that battery innovation should be assessed as a set of trade-offs. A lower-cost material may require more material per unit of stored energy; a higher-performing cell may be harder to manufacture consistently. The relevant strategic question is whether the complete product becomes more attractive at an acceptable cost. This report therefore treats R&D expenditure as an input and avoids using it as a direct measure of successful innovation.

3.2 Adoption and charging as complementary capabilities

Study 4 - Stockkamp et al. (2021). A systematic review of 49 studies brings together factors associated with electric-vehicle adoption. It highlights the interaction of product attributes, consumer characteristics, and the surrounding environment. The relevance to Tesla is that technical features do not translate into demand independently of price and everyday convenience. The reviewed studies differ in location and design, so their findings cannot be reduced to one universal customer preference. The underlying literature also predates the most recent expansion of the market.

Study 5 - Hardman and Tal (2021). Using repeated questionnaire surveys of California owners, the authors investigate why some households discontinue plug-in vehicle ownership. Charging convenience and access to suitable home charging help explain the pattern. This shifts attention from winning a first sale to supporting continued use. Tesla's charging and service experience may therefore influence retention. California's early adopters are not representative of all countries, and observed associations do not establish that charging access alone causes discontinuance.

Study 6 - Springel (2021). Norwegian vehicle-registry data and a two-sided market model show that vehicle demand and charging provision interact. The study finds that the relative effectiveness of purchase and station subsidies changes as spending increases. For Tesla, this provides a mechanism through which infrastructure can support vehicle adoption and through which a larger user base can support infrastructure. The results concern a particular policy setting; they do not prove that Tesla's network is permanently protected from competition.

Study 7 - Powell et al. (2022). A charging-demand and electricity-dispatch model examines the US Western Interconnection under deeper EV adoption. It finds that charging location and timing influence grid requirements; daytime charging can improve outcomes under the modeled system. For Tesla, charging expansion should consider electricity availability and site economics alongside coverage. The study models future grid conditions rather than Tesla's own charging profits, and its preferred timing cannot be transferred unchanged to every electricity system.

Interpretation

The four studies make ownership convenience a strategic variable. A strong vehicle can lose appeal if charging is difficult, expensive, or unreliable. Conversely, a well-operated network can improve a familiar product without changing its basic hardware. Infrastructure quality and customer retention therefore deserve attention alongside new technical features.

3.3 Manufacturing, supply resilience, and network coverage

Study 8 - Alochet et al. (2023). Evidence from 19 automakers examines whether the architecture of battery electric vehicles is mirrored in production and sourcing. The authors find continued integration of important activities rather than a complete move to outsourced, modular production. This challenges the assumption that EV manufacturing becomes simple once the engine is removed. Tesla's integration may help coordination, but integration itself is not unique. The study captures an evolving industry and does not identify one universally superior factory structure.

Study 9 - Wang et al. (2023). An expanded integrated assessment model evaluates how higher critical-material prices could affect China's EV transition. The scenarios show that material-cost pressure can slow adoption, while recycling and battery innovation offer longer-term responses. This places limits on cost improvements achieved within a factory: external input prices still matter. For Tesla, supplier resilience and chemistry choices are relevant to competitiveness. The outcomes are scenario estimates for China, not predictions of Tesla's margins.

Study 10 - Hanig et al. (2025). A geographic analysis develops a consecutive-coverage measure for US charging access. It finds that highway fast-charger deployment can connect many counties while leaving some rural gaps. The contribution is to assess usable travel coverage rather than simply count plugs. This offers a practical way to assess Tesla's network advantage. Coverage alone, however, does not establish reliability, waiting time, or profitability, and national results cannot be assumed to describe every Tesla site.

Synthesis and critical assessment

The literature explains several routes from innovation to customer value: improved batteries, coordination of production, easier charging, and more resilient supply. It also shows why those routes are conditional. Battery progress can spread to rivals, integration can become costly, and infrastructure can remain inconvenient even when total capacity rises. The studies complement one another, but they measure different outcomes and should not be pooled as if they estimate a common effect.

Most of this evidence concerns the EV sector or a particular national market. Its application to Tesla is an analytical interpretation, not a finding made by each author. This distinction is central to the report: published research identifies plausible mechanisms, while Tesla's financial series is used to examine whether the expected commercial benefits are visible at the company level.

4. NEED OF THE STUDY AND RESEARCH GAP

4.1 Need of the study

Tesla is often assessed through product announcements, technical specifications, or expectations about future businesses. Those perspectives can overlook the practical question of whether innovation produces an enduring return. Managers need to decide which projects deserve capital, customers need reliable products, and researchers need measures that distinguish investment from results. A study that connects technology to financial and operational evidence is therefore useful even when it cannot establish causality.

4.2 Research gap

Within the ten studies reviewed, evidence is distributed across battery costs, manufacturing organisation, customer adoption, and charging systems. They do not jointly test Tesla's 2019-2025 financial performance against these mechanisms. This is a bounded gap in the selected evidence, not a claim that no previous Tesla study exists. The present report connects that literature with a transparent annual dataset, a regulatory-credit sensitivity, and carefully qualified comparisons.

A second gap concerns measurement. Innovation is sometimes represented by R&D spending, while competitive advantage is inferred from revenue growth. Neither measure is sufficient on its own. Expenditure can increase before useful output appears; revenue can grow while margins decline. This study uses several indicators and treats their disagreement as evidence requiring explanation.

4.3 Theoretical basis

The resource-based view asks whether resources are valuable and difficult to reproduce or substitute (Barney, 1991). VRIO adds an explicit check on whether the organisation can capture their value. Here it is used as a diagnostic matrix, with judgments shown rather than presented as measured scores. Dynamic capabilities focus attention on sensing opportunities, committing resources, and changing operations as conditions shift (Teece, 2007).

Porter's (2008) five forces places those capabilities within an industry: rivals, customers, suppliers, new entrants, and substitutes affect how much value the firm can retain. These perspectives are complementary. A technically strong capability can still face price pressure, while a useful market position can weaken if the firm fails to adapt.

4.4 Analytical proposition

Innovation inputs may improve products and processes; those improvements may increase willingness to pay, reduce costs, or support repeat use. Competitive advantage emerges only if Tesla captures the benefit and rivals cannot readily neutralise it. The report evaluates this chain without assuming that every link has already been demonstrated.

5. RESEARCH METHODOLOGY

5.1 Objectives and research questions

Objective	Question addressed
1. Examine financial trends	How did revenue, gross margin, and R&D intensity change during 2019-2025?
2. Evaluate innovation mechanisms	Which capabilities could create value, and what limits their defensibility?
3. Place Tesla in competitive context	What can BYD and Toyota reveal once reporting differences are retained?
4. Explore association	Does the R&D-margin pattern remain consistent under timing and sensitivity checks?
5. Develop recommendations	Which actions follow from the evidence and its limitations?

5.2 Research design

The study is descriptive and analytical, using a longitudinal Tesla case with two contextual comparators. It combines secondary quantitative data with qualitative interpretation. No customer survey, employee interview, experiment, or causal regression was conducted. The design is appropriate for examining patterns and mechanisms, but it cannot isolate the financial return on an individual technology.

5.3 Sample population size and sampling technique

The conceptual population consists of vehicle manufacturers competing in electrification and their publicly available business evidence. Tesla is selected purposively because innovation is central to its competitive position. BYD and Toyota are selected for contrasting business configurations. The company sample is therefore three cases, while the core quantitative sample is seven Tesla company-year observations, a complete enumeration of 2019-2025 within the chosen window.

The literature sample contains ten purposively selected peer-reviewed studies published in 2021-2025. Publisher and institutional sources were searched using combinations of electric vehicle, battery cost, charging infrastructure, adoption, manufacturing, and competitive advantage. Inclusion required a clear method and relevance to an innovation mechanism. This is a focused narrative review, not an exhaustive systematic review or a representative sample of all research.

5.4 Variables and analytical tools

Measure	Definition and role
Revenue and gross profit	Reported USD millions; describe scale and gross earnings.
Gross margin	100 × gross profit / revenue; an imperfect performance proxy.
R&D intensity	100 × expensed R&D / revenue; an innovation-input proxy.
Credit sensitivity	100 × (gross profit - credits) / (revenue - credits).
Annual growth and CAGR	Annual percentage change; CAGR uses six intervals between 2019 and 2025.
Qualitative capabilities	Software, batteries, production, charging, and energy; coded using VRIO and five forces.

Descriptive trends, ratio calculations, and charts are combined with Pearson and Spearman correlations. A one-year lag and leave-one-year-out calculations test how sensitive the pattern is to timing and individual observations. With seven years, these are exploratory descriptions. No significance threshold is used to declare a causal effect or validate a hypothesis.

The independent recalculation uses the reported annual inputs before rounding. Ratios displayed in the report are rounded to two decimal places, while correlations are rounded to three decimals. The credit-adjustment formula removes the same revenue item from both the numerator and denominator.

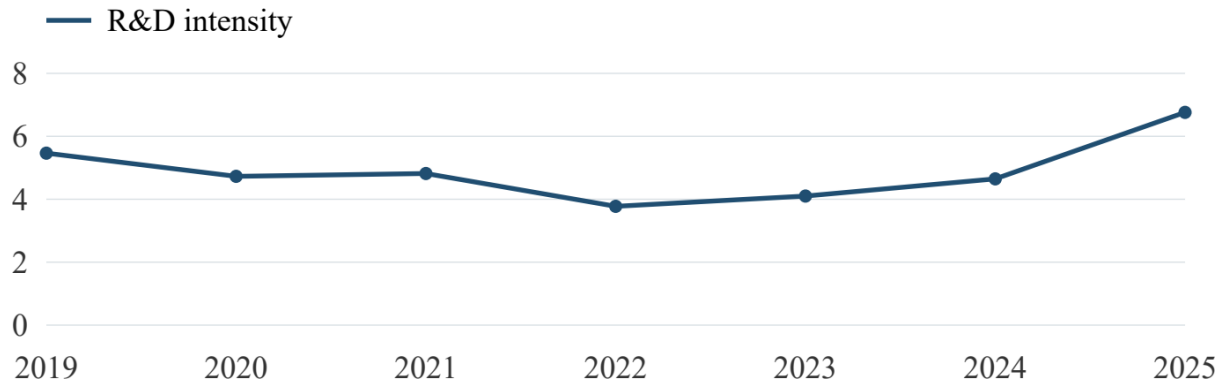


Figure 1: R&D intensity, percent of revenue. This chart illustrates the innovation-input variable; calculated from the reported annual data.

5.5 Data sources and verification

The supplied Tesla notebook and CSV archive provide the starting dataset. Calculations are independently recomputed. Tesla's 2020, 2022, and 2025 annual filings provide overlapping annual figures; BYD's investor-relations summary and Toyota's financial-results presentation provide comparator context. The IEA supplies industry background. Academic sources are used to assess mechanisms rather than to fill gaps in company accounts.

Figures retain their original currency and period. Toyota's R&D activity expenditure is explicitly distinguished from an income-statement R&D cost. Unverified BYD R&D inputs in the supplied files are not used in a comparative ranking. Promotional targets are not treated as realised cost savings.

5.6 Limitations

Seven observations offer little statistical power and may be serially dependent. Prices, volumes, product mix, input costs, regulation, and accounting changes can affect the ratios. Gross margin excludes R&D expense itself, although development may influence later production economics. Public disclosures do not identify project-level returns or customer willingness to pay. Strategic ratings remain reasoned judgments, and financial associations cannot establish causation.

The analysis distinguishes reported figures, calculated measures, and qualitative judgments. The original annual inputs are reproduced in the analysis section, and the calculation examples are included in the appendix. The pie charts use the same consolidated totals as the financial table.

6. DATA ANALYSIS AND INTERPRETATION

6.1 How Tesla innovations can create advantage

Software and over-the-air updates. Tesla develops much of its vehicle software internally and delivers remote updates and paid options (Tesla, 2026). The mechanism is continued product improvement after purchase. The strategic test is whether updates deliver useful, dependable functions that customers value, rather than merely increasing the number of available features. Software revenue is not separately isolated in this dataset.

Battery systems and manufacturing. Battery innovation includes cell design, pack integration, thermal management, and the process used to manufacture them. Tesla's 4680 programme is one example identified in the supplied notebook. Its proposed benefits cannot be accepted as achieved savings without production evidence. Larger cells, fewer parts, or more automation create an advantage only if yield, reliability, and total cost improve. The battery literature supports measuring these trade-offs rather than choosing a technology by headline specifications (Mauler et al., 2021; Yao et al., 2025).

Supercharging and network access. Tesla provides Supercharger access to eligible vehicles from other brands (Tesla, n.d.). This broadens the potential customer base for charging. The value mechanism combines convenient travel, network utilisation, and service revenue. A network can be valuable without being exclusive; its defensibility may depend increasingly on location, uptime, payment simplicity, and operating cost.

Driver assistance and AI. FSD (Supervised) and broader AI projects offer possible future differentiation, but a supervised product should not be described as unrestricted autonomous driving. A large data supply is an input, not proof of superior outcomes. Independent measures of reliability, safety, customer adoption, and economics are needed before assigning a durable advantage.

Powerwall and Megapack. Stationary storage applies related battery and control capabilities to a different customer need. It can widen the commercial use of technical knowledge. The financial analysis on page 18 examines the observed contribution; it does not assume that the economics of a stationary installation are the same as those of a vehicle.

The mechanisms above are interpretations. Product existence, technical promise, and demonstrated economic advantage are treated as separate questions.

6.2 Growth and margin performance

Year	Revenue	R&D	Gross profit	Credits
2019	24,578	1,343	4,069	594
2020	31,536	1,491	6,630	1,580
2021	53,823	2,593	13,606	1,465
2022	81,462	3,075	20,853	1,776
2023	96,773	3,969	17,660	1,790
2024	97,690	4,540	17,450	2,763
2025	94,827	6,411	17,094	1,993

Table 2: Tesla reported annual data, USD millions

Sources: supplied dataset; Tesla (2021, 2023, 2026), statements of operations and automotive-credit disclosures.

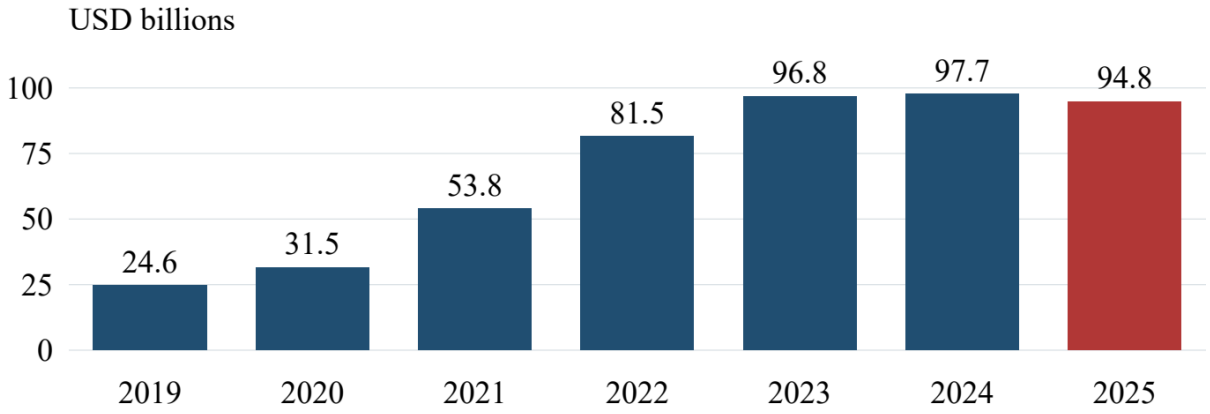


Figure 2: Tesla revenue, 2019-2025, USD billions

Revenue grew at a calculated 25.24% compound annual rate between 2019 and 2025. The path was uneven: rapid expansion through 2022 was followed by slower growth and a 2.93% decline in 2025. Gross margin peaked at 25.60% in 2022 and fell to 18.03% in 2025, a reduction of 7.57 percentage points.

The strongest inference is that greater scale did not preserve the earlier margin level. This does not demonstrate technological failure; lower prices, product mix, and production costs can change the share of value Tesla retains. It does weaken the assumption that a growing technology business automatically has a strengthening competitive advantage.

6.3 R&D investment and observable returns

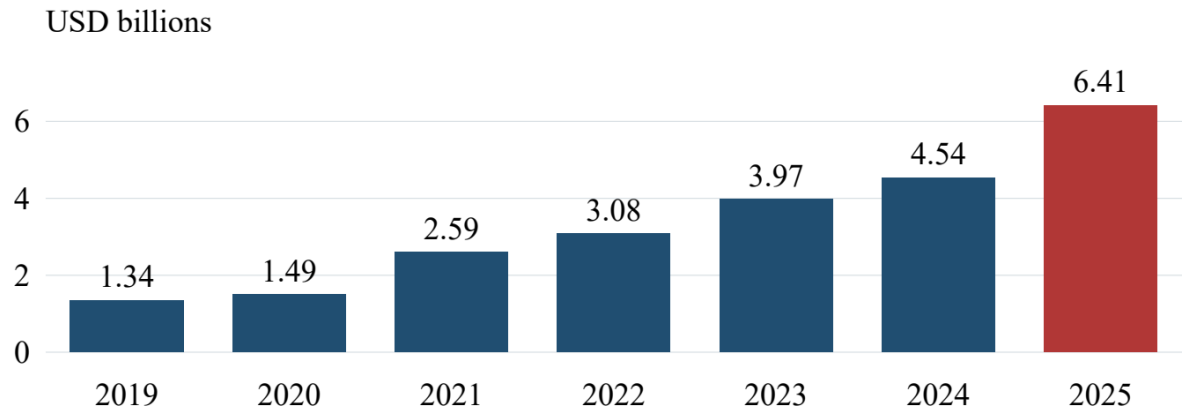


Figure 3: Tesla R&D expenditure, 2019-2025, USD billions

R&D expenditure grew at a calculated 29.76% compound annual rate, faster than revenue. In 2025 it increased by 41.21%, taking intensity to 6.76%. The ratio reflects both the spending increase and the revenue decline; it is not a pure measure of research productivity.

Table 3: Exploratory association and robustness checks

Comparison	n	Result
R&D intensity and same-year gross margin	7	Pearson r = -0.484; Spearman = -0.500
R&D intensity and next-year gross margin	6	Pearson r = 0.492
Same-year, each year omitted in turn	6	r ranges from -0.636 to -0.252
Approximate Fisher interval for same-year r	7	95% interval: -0.91 to 0.42

The negative same-year association cannot support a claim that R&D harms performance. It becomes positive when spending intensity is paired with the following year's margin. That reversal, the wide interval, and the small sample make a strong statistical conclusion inappropriate. The Fisher interval additionally assumes conditions that may not hold for an annual time series.

R&D is recorded below gross profit in Tesla's statements, so it does not mechanically reduce gross margin. The relevant question is whether it improves later products, costs, or revenue. This dataset does not observe project outputs or the time needed to commercialise them. The appropriate conclusion is uncertainty about the conversion of investment into returns.

6.4 Margin trends and regulatory credits

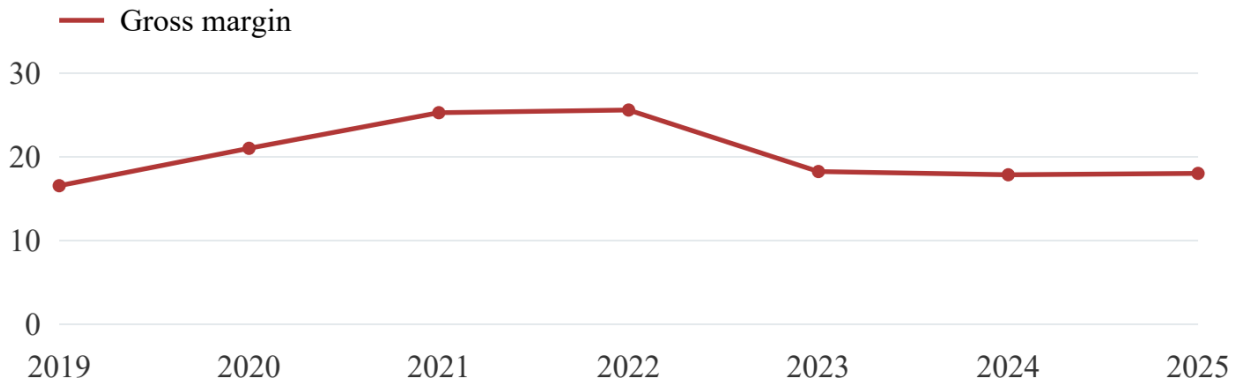


Figure 4: Consolidated gross margin, percent

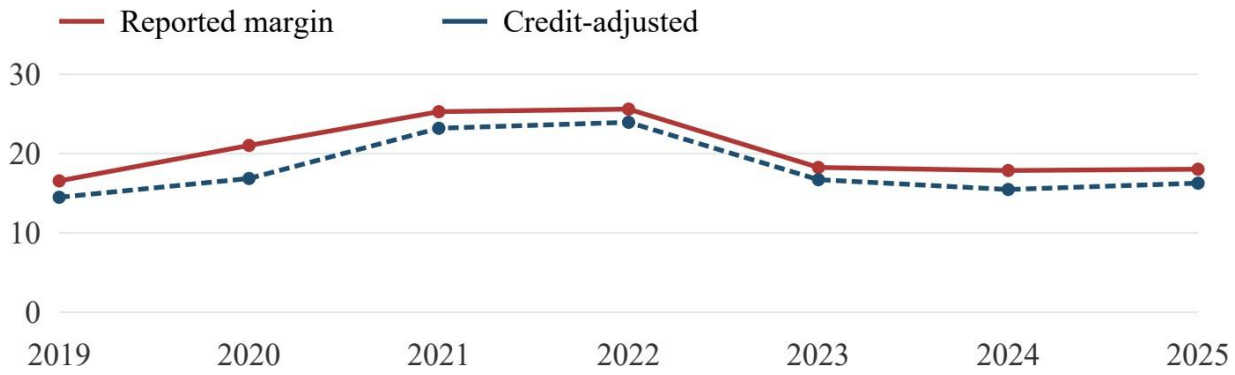


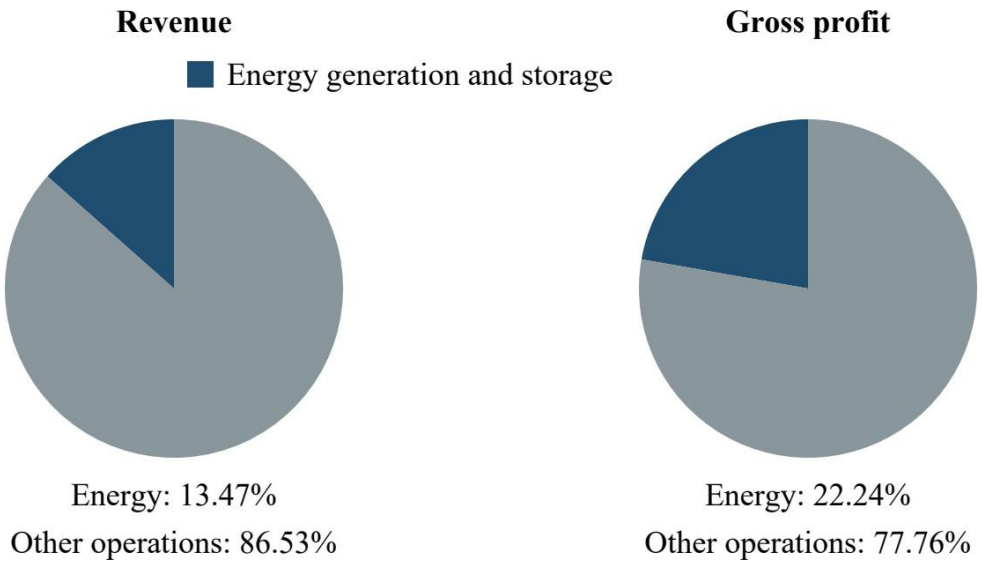
Figure 5: Reported margin and credit-adjusted sensitivity, percent

Year	Reported margin	Adjusted sensitivity	Credits / gross profit
2019	16.56%	14.49%	14.60%
2022	25.60%	23.94%	8.52%
2024	17.86%	15.47%	15.83%
2025	18.03%	16.27%	11.66%

Table 4: Selected margins and credit contribution

The sensitivity removes credits from both gross profit and revenue, assuming negligible incremental credit cost. It is a consolidated analytical measure, not Tesla's reported automotive margin. The 2025 result is 16.27%, compared with 18.03% reported. The adjusted result improved from 15.47% in 2024, but remained well below 23.94% in 2022. Credits therefore matter without fully explaining the broader margin pattern.

6.5 Energy diversification and business mix



Figures 6 and 7: Energy generation and storage share of revenue and gross profit, 2025

Energy generation and storage revenue reached $12.77 billion in 2025, with a reported segment gross margin of 29.8% (Tesla, 2026). Calculated from segment revenue and cost, its $3.80 billion gross profit represented about 22.24% of consolidated gross profit, versus 13.47% of revenue. This indicates that the business contributed disproportionately at the gross-profit level.

Energy storage is consequently relevant to the competitive-advantage question, rather than a minor background activity. However, segment gross margin is not a complete return-on-investment measure: it does not allocate every corporate expense or establish cash returns on new capacity. Diversification creates options, but those options still need commercial discipline.

The pie charts compare two different denominators. Energy generation and storage accounts for 13.47% of revenue and 22.24% of gross profit. Other operations include automotive activities and services. The charts show the relative contribution of the energy business; they do not measure net profit or its return on invested capital.

6.6 Competitive benchmarking with comparable definitions

Company and period	Reported inputs	Calculated or reported ratio
Tesla Jan-Dec 2025	Revenue USD 94,827m; gross profit USD 17,094m; R&D expense USD 6,411m.	Gross margin 18.03%; R&D expense / revenue 6.76%.
BYD Group Jan-Dec 2025	Revenue RMB 803,964.958m; gross profit RMB 142,659.797m.	Gross margin 17.74%.
Toyota Apr 2025-Mar 2026	Revenue JPY 50,684.9bn; operating income JPY 3,766.2bn; R&D activity expenditure JPY 1,522.8bn.	Operating margin 7.4% reported; R&D activity / revenue 3.00%.

Table 5: Contextual comparison; original currencies retained

Sources: Tesla (2026), BYD Company Limited (n.d.), and Toyota Motor Corporation (2026)

Tesla and BYD's consolidated gross margins are close in this snapshot, separated by approximately 0.28 percentage points. This is not evidence of equal vehicle-level efficiency: BYD's group includes other operations, and product mix and accounting differ. BYD Electronic is not substituted for BYD Group. Absolute revenues are not ranked across currencies.

Toyota's operating margin is shown separately because it deducts operating expenses that gross margin excludes. Its R&D measure covers activity-related expenditure and explicitly does not conform to R&D costs in the consolidated income statement. The 3.00% ratio therefore cannot be used to conclude that Toyota is less innovative or that Tesla's research is more productive.

The strategic lesson is that Tesla faces capable competitors using different combinations of technology and scale. A useful benchmark asks what customers receive and what it costs to deliver that outcome. A future matched comparison should control for vehicle segment, geography, battery specification, and warranty obligations before attributing a cost or margin difference to innovation.

6.7 Strategic assessment and key findings

Capability	V / R / I / O	Interpretation
Software and vehicle integration	Yes / Partial / Partial / Yes	Useful differentiation; exclusive returns not established.
Charging network and service	Yes / Partial / Partial / Yes	Coverage and operation matter; broader access can increase use but reduce exclusivity.
Battery and manufacturing know-how	Yes / Partial / Partial / Partial	Execution can create gains; company-level data do not isolate their returns.
Fleet learning for AI	Potential / Unclear / Unclear / Partial	More data alone do not prove better safety, performance, or profit.

Table 6: VRIO assessment. V = valuable; R = rare; I = hard to imitate; O = organised to capture value. Ratings are analytical judgments.

Evidence basis: Tesla (2026, n.d.); Alochet et al. (2023); Springel (2021); Yao et al. (2025).

The matrix supports a conditional advantage rather than an unqualified permanent moat. Integration can make several activities work better together, but the literature shows that rivals also integrate important systems. Opening charging access to other brands creates a trade-off between network revenue and vehicle exclusivity (Tesla, n.d.).

Five forces and SWOT synthesis

Rivalry: high. Alternative manufacturers and converging capabilities pressure differentiation. **Buyer power: moderate to high.** Buyers can compare features and costs, although individual consumers have limited negotiating power. **Supplier power: moderate to high.** Critical materials and specialised inputs create exposure. **New entrants: moderate.** Capital and service requirements restrain entry. **Substitutes: high.** Hybrids and conventional vehicles remain alternatives (IEA, 2025; Wang et al., 2023; Porter, 2008).

In SWOT terms, coordinated capabilities are a strength; compressed margins and uncertain project returns are weaknesses. Energy storage and better charging services offer opportunities, while imitation and input-cost pressure are threats. The practical response is to strengthen measurable customer benefits and execution, rather than assume that every new feature increases defensibility.

7. RECOMMENDATIONS AND IMPLICATIONS

1. Make innovation spending accountable to milestones. Within six months, finance and engineering should review major projects against technical readiness, unit economics, and customer adoption. Continue funding in stages; revise projects that repeatedly miss agreed milestones. Higher expenditure alone should not be treated as success.

2. Prioritise dependable ownership. Charging and service teams should track successful charging sessions, peak waiting time, repair turnaround, and repeat purchase intentions. Use site-level coverage analysis to guide expansion over the following 12 months. This addresses the convenience mechanisms identified in the literature.

3. Test manufacturing gains before scaling them. Operations should evaluate yield, scrap, warranty cost, and capital per unit before expanding new battery or production processes. Compare results with feasible supplier alternatives. A technically ambitious process is valuable only if it performs consistently at commercial volume.

4. Allocate capital across vehicles and energy on comparable returns. Review cash conversion, warranty exposure, and return on invested capital quarterly. Report credit-adjusted sensitivity alongside reported results internally. These measures would help distinguish operating progress from revenue mix and policy-related earnings.

For managers, the implication is to connect engineering targets with ownership and financial outcomes. For researchers, the next step is a larger quarterly or multi-firm dataset with controls for price, mix, and production changes. Customer evidence would add a missing view of willingness to pay and retention.

7.1 Implementation priorities

A practical sequence is to establish project milestones first, then use charging and service measures to identify the most immediate ownership problems. Manufacturing expansion can be assessed against consistent unit-economics criteria, while vehicle and energy investments are reviewed using comparable cash-return measures. Each review should distinguish a technical achievement from a demonstrated commercial return.

8. CONCLUSION

Tesla's experience shows that innovation can expand a business without preserving its earlier margins. The seven-year evidence combines substantial growth with a marked decline in gross margin after 2022. R&D intensity alone does not explain that pattern, and the exploratory correlations are too unstable to support causal claims.

The most credible source of advantage is coordination across products, software, production, charging, and energy applications. Its durability depends on reliable execution and continued customer value as competitors improve. Tesla therefore needs to demonstrate how technical progress becomes better ownership, lower economic cost, and sustainable returns. The overall outcome is a qualified finding of competitive strength, with its persistence still requiring evidence.

8.1 Overall outcome

The evidence supports a conditional competitive advantage. Tesla possesses useful complementary capabilities, but their economic value depends on how reliably the business turns them into customer benefits and returns. Future research should test this relationship with more observations and better measures of product outcomes, customer retention, and project-level economics.

9. BIBLIOGRAPHY

Alochet, M., MacDuffie, J. P., & Midler, C. (2023). Mirroring in production? Early evidence from the scale-up of battery electric vehicles (BEVs). *Industrial and Corporate Change, 32*(1), 61-111. https://doi.org/10.1093/icc/dtac028

Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management, 17*(1), 99-120. https://doi.org/10.1177/014920639101700108

BYD Company Limited. (n.d.). *Investor relations* [Financial summary]. Retrieved October 14, 2026, from https://www.bydglobal.com/cn/investor-relations?scroll=true

Hanig, L., Ledna, C., Nock, D., Harper, C. D., Yip, A., Wood, E., & Spurlock, C. A. (2025). Finding gaps in the national electric vehicle charging station coverage of the United States. *Nature Communications, 16*, Article 561. https://doi.org/10.1038/s41467-024-55696-8

Hardman, S., & Tal, G. (2021). Understanding discontinuance among California's electric vehicle owners. *Nature Energy, 6*, 538-545. https://doi.org/10.1038/s41560-021-00814-9

International Energy Agency. (2025). *Global EV outlook 2025.* https://www.iea.org/reports/global-ev-outlook-2025

Mauler, L., Duffner, F., Zeier, W. G., & Leker, J. (2021). Battery cost forecasting: A review of methods and results with an outlook to 2050. *Energy & Environmental Science, 14*(9), 4712-4739. https://doi.org/10.1039/D1EE01530C

Porter, M. E. (2008). The five competitive forces that shape strategy. *Harvard Business Review, 86*(1), 78-93. https://hbr.org/2008/01/the-five-competitive-forces-that-shape-strategy

Powell, S., Cezar, G. V., Min, L., Azevedo, I. M. L., & Rajagopal, R. (2022). Charging infrastructure access and operation to reduce the grid impacts of deep electric vehicle adoption. *Nature Energy, 7*, 932-945. https://doi.org/10.1038/s41560-022-01105-7

Springel, K. (2021). Network externality and subsidy structure in two-sided markets: Evidence from electric vehicle incentives. *American Economic Journal: Economic Policy, 13*(4), 393-432. https://doi.org/10.1257/pol.20190131

Stockkamp, C., Schäfer, J., Millemann, J. A., & Heidenreich, S. (2021). Identifying factors associated with consumers' adoption of e-mobility - A systematic literature review. *Sustainability, 13*(19), Article 10975. https://doi.org/10.3390/su131910975

Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal, 28*(13), 1319-1350. https://doi.org/10.1002/smj.640

Tesla, Inc. (n.d.). *Supercharging other EVs.* Retrieved October 14, 2026, from https://www.tesla.com/support/charging/supercharging-other-evs

Tesla, Inc. (2021). *Annual report on Form 10-K for the year ended December 31, 2020.* U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1318605/000156459021004599/tsla-10k_20201231.htm

Tesla, Inc. (2023). *Annual report on Form 10-K for the year ended December 31, 2022.* U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1318605/000095017023001409/tsla-20221231.htm

Tesla, Inc. (2026). *Annual report on Form 10-K for the year ended December 31, 2025.* U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1318605/000162828026003952/tsla-20251231.htm

Toyota Motor Corporation. (2026, May 8). *FY2026 financial results* [Presentation]. https://global.toyota/pages/global_toyota/ir/financial-results/2026_4q_presentation_2_en.pdf

Wang, H., Feng, K., Wang, P., Yang, Y., Sun, L., Yang, F., Chen, W.-Q., Zhang, Y., & Li, J. (2023). China's electric vehicle and climate ambitions jeopardized by surging critical material prices. *Nature Communications, 14*, Article 1246. https://doi.org/10.1038/s41467-023-36957-4

Yao, A., Benson, S. M., & Chueh, W. C. (2025). Critically assessing sodium-ion technology roadmaps and scenarios for techno-economic competitiveness against lithium-ion batteries. *Nature Energy, 10*, 404-416. https://doi.org/10.1038/s41560-024-01701-9

Ziegler, M. S., & Trancik, J. E. (2021). Re-examining rates of lithium-ion battery technology improvement and cost decline. *Energy & Environmental Science, 14*(4), 1635-1651. https://doi.org/10.1039/D0EE02681F

10. APPENDIX

Appendix A - Reproducibility and supporting calculations

Year	Gross margin	R&D intensity	Adjusted margin
2019	16.56%	5.46%	14.49%
2020	21.02%	4.73%	16.86%
2021	25.28%	4.82%	23.19%
2022	25.60%	3.77%	23.94%
2023	18.25%	4.10%	16.71%
2024	17.86%	4.65%	15.47%
2025	18.03%	6.76%	16.27%

Worked example, 2025. Gross margin = 17,094 / 94,827 × 100 = 18.03%. R&D intensity = 6,411 / 94,827 × 100 = 6.76%. Credit-adjusted sensitivity = (17,094 - 1,993) / (94,827 - 1,993) × 100 = 16.27%. Revenue CAGR = $[(94{,}827 / 24{,}578)^{1/6} - 1] \times 100 = 25.24\%$. Calculations use unrounded inputs.

Appendix B - Source and quality record

The supplied files were *Tesla_Innovation_Study.ipynb* and *tesla_study_outputs.zip*. The report uses their annual data as a starting point and independently recomputes ratios and correlations. Tesla's 2020 filing covers 2019-2020; its 2022 filing covers 2021-2022; its 2025 filing covers 2023-2025. The notebook's verification labels are not accepted as evidence by themselves.

The report retains the correct credit-adjustment denominator, distinguishes gross and operating margins, and excludes an unverified comparative R&D ranking. Toyota's R&D definition is taken from the footnote to its results presentation. Battery-development targets are not represented as achieved savings. Strategic assessments are explicitly identified as judgments.

Appendix C - Primary research status

No questionnaire was administered and no respondents were sampled. A questionnaire is therefore not presented as completed research. Future work could survey owners about charging access, service experience, software usefulness, and repurchase intentions, with informed consent and an independently justified sampling plan.